

InMoov — Right Hand + Forearm

Professional 3-Phase Print Checklist • for parallel print + assembly • A4 • 24 parts

FILAMENT & PRINTER REQUIREMENTS

Material	PLA+ <i>recommended</i> • PLA or PETG acceptable. △ WARNING: ABS only if your printer has an enclosure.
Color	WHITE (official InMoov standard)
Diameter	1.75 mm
Nozzle	0.4 mm
Bed temp	60 °C (PLA / PLA+) • 70–80 °C (PETG)
Nozzle temp	200–210 °C (PLA) • 230–240 °C (PETG)
Est. filament	~800 g – 1 kg total
Scale	100% — DO NOT RESCALE ANY PART

► FIRST PRINT — BEFORE ANYTHING ELSE: `CALIBRATOR.stl`

Download from inmoov.fr and **test-fit before printing any real parts**. If parts come out too tight, set **horizontal expansion** to **–0.15 mm** in your slicer and re-test.

PHASE 1 • FOREARM + WRIST BACKBONE

PRINT FIRST

"Print these first — start assembling the forearm immediately after"

Done	STL part file	Layer	Infill	Walls	Supp.	Raft
☐	1.robpart2V3.stl	0.25 mm	30%	2 mm	NO	YES
△ WARNING: PRINT FIRST — glues to rotawrist1. The square boss on the RotaWrist must align with the <i>single</i> hole of robpart2. Redrill the side holes with a 6 mm drill. Raft = anti-warp.						
☐	2.robpart3V3.stl	0.25 mm	30%	2 mm	NO	YES
Note: Pairs with robpart4 — glue these two together as a set. Raft = anti-warp.						
☐	3.robpart4V3.stl	0.25 mm	30%	2 mm	NO	YES
Note: Pairs with robpart3 — glue together. The servo bed spans robpart3/4/5. Raft = anti-warp.						
☐	4.robpart5V3.stl	0.25 mm	30%	2 mm	NO	YES
△ WARNING: Longest print, ~7 hrs+. Insert two 4 mm bolts into the printed cavities — heat the bolts with hot air if they don't fit. The servo bed seats here. Raft = anti-warp.						
☐	5.robcap3V1.stl	0.25 mm	30%	2 mm	NO	NO
☐	6.rotawrist1V3.stl	0.25 mm	30%	3 mm	YES	NO
△ WARNING: PRINT FIRST. Remove support with pliers after printing. Dry-fit against robpart2 BEFORE gluing — the glue face must align perfectly. Wrong orientation = broken build. See the photo at inmoov.fr/rotational-wrist/ .						
☐	7.rotawrist2.stl	0.15 mm	50%	3 mm	NO	NO
△ WARNING: Redrill with a 2.5 mm drill after printing. TIP: spray-paint it BLACK after printing (grease turns it yellow over time). Apply grease to the gear before assembly. 0.15 mm layer = best quality for the gear teeth.						
☐	8.rotawrist3V2.stl	0.25 mm	30%	3 mm	NO	NO
Note: Redrill with an 8 mm drill after printing. Mounts i2_WristLargeV2 on top.						
☐	9.RobServoBedV6.stl	0.2 mm	30%	2 mm	NO	NO
△ WARNING: Dry-fit inside the robpart3/4/5 shell before final assembly. File the servo-bed rails if needed for fit. Secure with 2 wood screws. Spans robpart3+4+5, not just one section.						
☐	10.RobCableFrontV3.stl	0.2 mm	30%	2 mm	NO	NO
☐	11.RobCableBackV3.stl	0.2 mm	30%	2 mm	NO	NO

Phase 1: 11 prints.

► ASSEMBLY UNLOCKED

- Glue robpart2 + robpart5 together.
- Glue robpart3 + robpart4 together.
- Dry-fit rotawrist1 on robpart2, then glue — **check orientation!**
- Insert the **MG996R** wrist servo (180° servo).
- Fit the servo bed into the robpart3/4/5 shell.

"Print while assembling Phase 1 — start finger assembly after"

Done	STL part file	Layer	Infill	Walls	Supp.	Raft
☐	12. i2_FingersX5V2.stl	0.25 mm	30%	2 mm	NO	NO
Note: Contains all 5 fingers in one file (×5 print). After printing, redrill ALL finger hinge holes with a 3 mm drill bit . Assembly order: Middle first , then Index, Ring, Pinky, Thumb last .						
☐	13. i2_FingersMoldX5V3.stl	0.25 mm	30%	2 mm	NO	NO
Note: Mold used to cast silicone fingertips (optional). Run fishing line ~30 cm through the spring hole of each finger section.						
☐	14. i2_WristLargeV2.stl	0.25 mm	30%	2 mm	NO	NO
Note: Redrill with an M8 drill bit. Mounts to rotawrist3 with an M8 printed bolt + mini clamp on the bolt groove. PTFE tubes (ID 1.5 mm × OD 2.5 mm) run through the main-gear Teflon pipes. PTFE lengths: Thumb 335 mm • Index 300 mm • Middle 300 mm • Ring 300 mm • Pinky 360 mm.						
☐	15. i2_WristGearV1.stl	0.15 mm	50%	3 mm	NO	NO
Note: This is the tendon pass-through gear, <i>not</i> the rotation gear. PTFE tubes run through this gear's Teflon-pipe holes. 0.15 mm = best quality (gear part).						
☐	16. i2_PulleyX5V1.stl	0.15 mm	50%	3 mm	NO	NO
Note: Hand-side pulley (redirects the tendon at the hand). DIFFERENT from servo-pulleyX5 — both are required. 0.15 mm = best quality.						
☐	17. RobRingV3.stl	0.2 mm	30%	2 mm	NO	NO
☐	18. TensionerRightV1.stl	0.2 mm	30%	2 mm	NO	NO
△ WARNING: RIGHT-hand version only — do NOT substitute a Left version. Screws onto the servo bed.						
☐	19. servo-pulleyX5.stl	0.15 mm	50%	3 mm	NO	NO
Note: Mounts on the servo output horns inside the servo bed (×5, one per servo). Redrill with a 2 mm drill after printing. DIFFERENT from i2_PulleyX5V1 — both are required. 0.15 mm = best quality (takes direct servo load).						

Phase 2: 8 prints.

► ASSEMBLY UNLOCKED

- Mount i2_WristLargeV2 to rotawrist3 with the M8 bolt; set the mini clamp on the bolt groove.
- Assemble the fingers to WristLarge with **M3×16 mm** screws (order: Middle, Index, Ring, Pinky, Thumb).
- Install 5× **MG995** finger servos in the servo bed — **set each servo to 150° before mounting**.
- Run PTFE tubes through the WristGear Teflon pipes.
- Run PTFE tubes through the servo-bed holes to the servo shafts.

"Hand is functional without these — print while doing Phase 2 assembly"

Done	STL part file	Layer	Infill	Walls	Supp.	Raft
☐	20. i2_CoverFingerV3.stl	0.25 mm	30%	2 mm	YES	If needed
TIP: Print STANDING UPRIGHT for the best surface result (not flat).						
☐	21. i2_FingersTipX5V2.stl	0.25 mm	30%	2 mm	YES	If needed
Note: Glue the tips onto the finger sections AFTER assembly. Do <i>not</i> glue them on before assembly — they look wrong in the closed position.						
☐	22. i2_HandCoverV1.stl	0.25 mm	30%	2 mm	YES	If needed
Note: Before screwing, tap the holes with an M3 tap for M3×16 mm screws.						
☐	23. i2_PalmCoverV2.stl	0.25 mm	30%	2 mm	YES	If needed
△ WARNING: When screwing the palm cover, set the ribbons neatly into the grooves. AVOID pinching ribbons — this kills the electronics.						
☐	24. i2_Adapter1109MG.stl	0.25 mm	30%	2 mm	YES	If needed
Note: Servo adapter for the 1109MG servo type.						

Phase 3: 5 prints.

► ASSEMBLY UNLOCKED

- Tap all cover holes with the M3 tap.
- Mount the HandCover and PalmCover.
- Route all ribbon cables neatly into the PalmCover grooves.
- Glue the FingerTips onto the finger sections.
- Final tensioning of the fishing-line tendons.

BUILD TOTALS & SIGN-OFF

Total STL files to print: 24 (Phase 1: 11 • Phase 2: 8 • Phase 3: 5)
Estimated total print time: 40–60 hours (varies by printer speed)
Phases: 3 — print + assemble in parallel
Assembly starts: after Phase 1 completes

Printer name: _____

Phase 1 started: _____ completed: _____

Phase 2 started: _____ completed: _____

Phase 3 started: _____ completed: _____

TOOLS & CONSUMABLES NEEDED FOR ASSEMBLY

Drills & taps

- ☐ 3 mm drill bit — finger hinge holes
- ☐ 6 mm drill bit — robpart2 side holes
- ☐ 8 mm drill bit — rotawrist3 + WristLarge
- ☐ 2 mm drill bit — servo-pulleyX5 holes
- ☐ 2.5 mm drill bit — rotawrist2 holes
- ☐ M3 tap — cover mounting holes

Adhesives & lubricant

- ☐ Epoxy (2-component) — rotawrist1 to robpart2
- ☐ Zap-A-Gap or epoxy — robpart shell joints
- ☐ Grease / lubricant — rotawrist gear

Hardware & consumables

- ☐ Fishing line ~30 cm ×5 — finger tendons
- ☐ PTFE tube ID 1.5 mm / OD 2.5 mm — actuation tubes
- ☐ M3×16 mm screws — fingers + covers
- ☐ Wood screws — servo-bed mounting

Hand tools

- ☐ Tweezers — running lines through fingers
- ☐ Sharp blade — clearing the hall-sensor exit hole
- ☐ Pliers — removing support from rotawrist1
- ☐ Hot-air source — seating 4 mm bolts in robpart5

Settings sourced from inmoov.fr/hand-and-forarm/, inmoov.fr/hand-i2/, inmoov.fr/rotational-wrist/ and community builders. Print-time estimate assumes ~50–60 mm/s; faster/slower printers will differ. Verify all redrill/tap operations on a test print before committing to a full set.